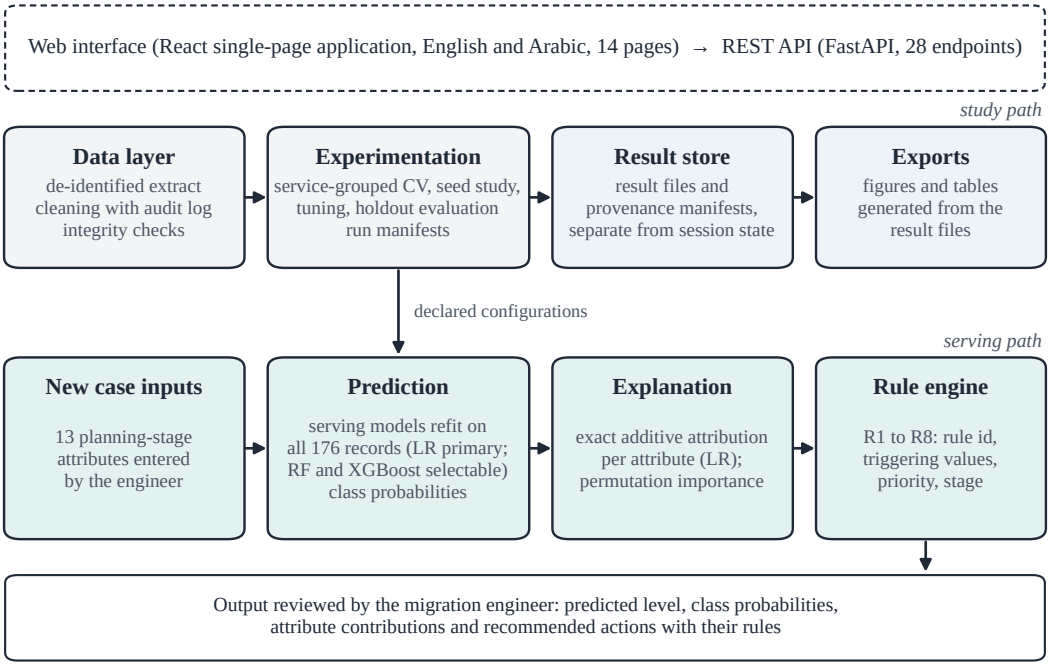
Web interface (React single-page application, English and Arabic, 14 pages) → REST API (FastAPI, 28 endpoints)
study path
Data layer
de-identified extract
cleaning with audit log
integrity checks
Experimentation
service-grouped CV, seed study,
tuning, holdout evaluation
run manifests
Result store
result files and
provenance manifests,
separate from session state
Exports
figures and tables
generated from the
result files
declared configurations
serving path
New case inputs
13 planning-stage
attributes entered
by the engineer
Prediction
serving models refit on
all 176 records (LR primary;
RF and XGBoost selectable)
class probabilities
Explanation
exact additive attribution
per attribute (LR);
permutation importance
Rule engine
R1 to R8: rule id,
triggering values,
priority, stage
Output reviewed by the migration engineer: predicted level, class probabilities,
attribute contributions and recommended actions with their rules